Dr. ANDREJ SVETLOŠÁK

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• https://andrejsvetlosak.github.io/ • EU Citizen, UK work and residence rights

Statistician and interdisciplinary Data Scientist, with focus on nonparametric statistics, clustering, unsupervised machine learning, customized statistical and interpretable AI solutions for industry, government and/or other involved parties. Experienced in communication and cooperation between academia, an established FTSE 250 asset management company, fintech startups, and governmental innovation hubs. Award holder for best teacher at the School of Business and Economics of the Freie Universität Berlin.

EDUCATION

The University of Edinburgh **Edinburgh, Scotland**
Doctor of Philosophy, Statistics **September 2017-July 2023**

- Thesis title "Novel Statistical Learning Approaches for Open Banking-Type Data".
- Industrial doctorate scholarship awarded by the Data Lab (an innovation centre funded by the SFC), The School of Mathematics and the Business School of the University of Edinburgh, and Inbest.ai.
- Focus on nonparametric statistics, unsupervised machine learning, clustering and market segmentation and developing methodology tailored for micro- and macroeconomic analysis of open banking-type data.
- Coordination of designated parts of the project, and communication of results between all involved parties.
- Organization of multiple events for participants from industry, academia, governmental organizations, and NGOs.

Freie Universität Berlin **Berlin, Germany**
Bachelor of Science, Economics (Grade 1.6) **2010-2013**

- Thesis on time inconsistent consumer behaviour and its welfare implications.
- Key coursework: empirical data analysis and statistics, game theory, environmental-, behavioural-, labour- micro- and macroeconomics.

PROFESSIONAL EXPERIENCE

The University of Edinburgh, School of Mathematics **Edinburgh, Scotland**
PDRA in Statistical Learning & Financial AI **April 2024 – April 2026**

- Developing a practical and methodological interface between statistical intelligence, interpretable machine learning and large language models, with applications practice-oriented to micro- and macroeconomic analysis.
- Engage as an expert in joint project of the University of Edinburgh (School of Mathematics, School of Informatics, Centre for Investing Innovation) and a FTSE 250 company.
- Contributing to project coordination and stakeholder engagement, ensuring effective communication across academia and industry.
- Manage and implement production-level AI software in multi-lingual environments (Python/R).

The University of Edinburgh, School of Mathematics **Edinburgh, Scotland**
Doctoral researcher **September 2017 – December 2023**

- Developed statistical and machine learning methods tailored for micro- and macroeconomic analysis of open banking-type data.
- Focus on nonparametric statistics and unsupervised machine learning. Mainly efficient clustering of large data using mixture models and game theory, nonparametric regression of time series.
- Engaged as an expert in a project between the University of Edinburgh (School of Mathematics and Business School), the Data Lab (funded by the SFC), and industry (Inbest.ai).
- Organized and led multi-stakeholder events for participants from industry, academia, governmental organizations, and NGOs.

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TEACHING AND OTHER WORK EXPERIENCE

- **Data Mining Tutor (Master Level)**
  Business School of the University of Edinburgh| 2017 – 2018
  Edinburgh, Scotland | Sector: Master-level University Education
- **Economics Tutor (Bachelor Level)**
  Freie Universität Berlin| 2011 –2017
  Berlin, Germany | Sector: Bachelor-level University Education
- **Statistician and Assistant of the Market Research Studio Director**
  Krämer Marktforschung | September 2015 - October 2015
  Berlin, Germany | Sector: Market Research
- **Student Assistant for Microeconometric Data Analyses to Prof. Dr. Steiner**
  Freie Universität Berlin| April 2012 – August 2012
  Berlin, Germany | Sector: Academic Research Assistant
- **Multinational Academy Látky (Deutsche Schüler Akademie)**
  Course Designer and Economics Lecturer | August 2012
  Látky, Slovakia |Sector: Education

PUBLICATIONS, CONFERENCE TALKS, & OTHER ACADEMIC ACTIVITIES

- Publications:
  - de Carvalho, M., Martos, G., & Svetlošák, A. (2025). A game-inspired algorithm for marginal and global clustering. *Pattern Recognition*, *160*, 111158.
  - Svetlošák, A. (2024). Novel statistical learning approaches for open banking-type data.
  - Svetlošák, A., de Carvalho, M., & Calabrese, R. (2023). Subject-to-group statistical comparison for open banking-type data. *Journal of the Operational Research Society*, *74*(3), 703-718.
- Talks:
  - Digital Transformation and Digital Economics: Principles and Practice (DXDE) 2024 (invited speaker).
  - The International Conference of the ERCIM WG on Computational and Methodological Statistics (2018, 2019, 2022, and 2025).
  - The 17-th conference of the International Federation of Classification Societies 2022.
  - The 7-th edition of the workshop on Models and Learning in Clustering and Classification 2022.
  - International Workshop on Statistical Modelling 2019 (poster).
  - Credit Scoring and Credit Control Conference XVI 2019 (poster).
  - Centre for Statistics Annual Conference 2019 (poster).
  - Financial Health Check Data Fest 2017 and 2018 (organizer and speaker).
- Other academic events and activities.
  - Co-organisor of Statistics Seminars of the University of Edinburgh.
  - Session chair in the Centre for Statistics Annual Conference 2024, 2025.
  - Reviewer of articles in the journal the *Journal Computational and Graphical Statistics* and *Plos One*.
  - Organisational assistant of the 2018 International Society for Bayesian Analysis conference,
  and Generative AI Models for Extreme Events 2025.

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SELECTION OF AWARDS & SCHOLARSHIPS

- Scholarship for an industrial doctorate awarded by the Data Lab, The University of Edinburgh and Inbest.ai. Link.
- Teaching award of the School of Economics and Business of the FU Berlin 2017. Link.
- Best in class in Economics. Link.
- DAAD prize for best foreign student at the FU Berlin. Link.
- Scholarship for university studies in Germany awarded by the German Academic Exchange Service (DAAD) – "Vollstipendium zur wissenschaftlichen Aus- und Fortbildung in Deutschland ".

LANGUAGES & PROGRAMING

Languages: Slovak (native), English (native level), German (native level).

Main programming language: R

- Development and application of nonparametric statistical algorithms, unsupervised learning methods, time series analysis models, and interpretable AI techniques.
- Visualisation via R Shiny and GGplot.

Secondary programming language: Python

- Development of software and implementation of methods from AI, machine learning, language models and statistics.
- Implementation and maintenance of shared code modules.
- Visualisation via Dash and Plotly.

Other software skills: LaTeX, SQL, Git, and Microsoft Office suite.

ACADEMIC REFERENCES

- Miguel de Carvalho, Professor & Chair of Statistical Data Science, University of Edinburgh, School of Mathematics, Miguel.deCarvalho@ed.ac.uk.
- Sotirios Sabanis, Professor & Personal Chair of Stochastic Analysis and Algorithms, Centre Director, Centre for Investing Innovation, University of Edinburgh, School of Mathematics, mailto:S.Sabanis@ed.ac.uk.
- Tiejun Ma, Professor & Personal Chair of Financial Computing (Risk Modelling), University of Edinburgh, School of Informatics, mailto:tiejun.ma@ed.ac.uk.